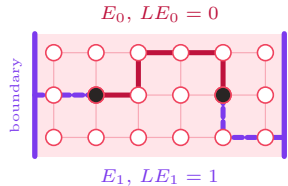
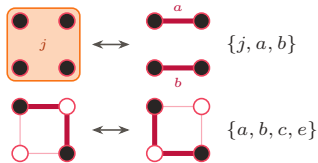


(a) Two-class search



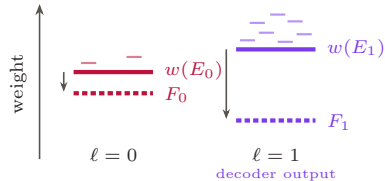
BP-OSD on $\begin{pmatrix} H \\ L \end{pmatrix} E = \begin{pmatrix} \sigma \\ \ell \end{pmatrix}$ for $\ell = 0$ and $\ell = 1$ gives a low-weight error in each logical class.

(b) Degeneracy moves



A move g adds faults whose detectors and observable flips cancel. It keeps σ and ℓ and changes the weight by Δw_g .

(c) Free-energy comparison



Each class is scored by $F_\ell = w(E_\ell) - \kappa \sum_g \ln(1 + e^{-\Delta w_g})$ (arrows). Short bars mark equivalent errors of slightly higher weight.